INORGANIC CHEMISTRY CUMULATIVE EXAM

December 3, 1994

Information about the chemistry of compounds is hidden in a variety of places. For example, careful observations of reactivities and product distribution can provide an abundance of useful information. It is the job of the research chemist to make the most from these observations. Below are three problems in which deductions must be made from observations. Answer all three, split your time evenly amongst them and be sure to write down each bit of information you can extract from the observations

1. The rate of reduction of $[Co(NH_3)_5(OH_2)]^{3+}$ by Cr(II) is seven orders of magnitude slower than the reduction of its conjugate base $[Co(NH_3)_5(OH)]^{2+}$ by Cr(II). For the corresponding reductions with $[Ru(NH_3)_6]^{2+}$ the two rates differ by less than a factor of 10. What do these observations suggest about the mechanisms?

2. The intermediate $[Fe(SCN)(OH_2)_5]^{2+}$ can be detected in the reaction (in water) of $Co(NCS)(NH_3)_5]^{2+}$ with Fe(II) to give Fe(III) and Co(II). What does this observation suggest about the mechanism?

3. When cobalt(II) salts are oxidized by air in a solution containing ammonia and sodium nitrite, a yellow solid, $[Co(NO_2)_3(NH_3)_3]$ can be isolated. In solution it displays a conductivity which is small enough to be attributed to impurities only. Werner treated it with HCl to give a complex which, after a series of further reactions, he identified as *trans*-$[CoCl_2(NH_3)_3(OH_2)]^+$. It required an entirely different route to prepare *cis*-$[CoCl_2(NH_3)_3(OH_2)]^+$. Is the yellow substance *fac* or *mer*? What assumptions must you make in order to arrive at your conclusion?